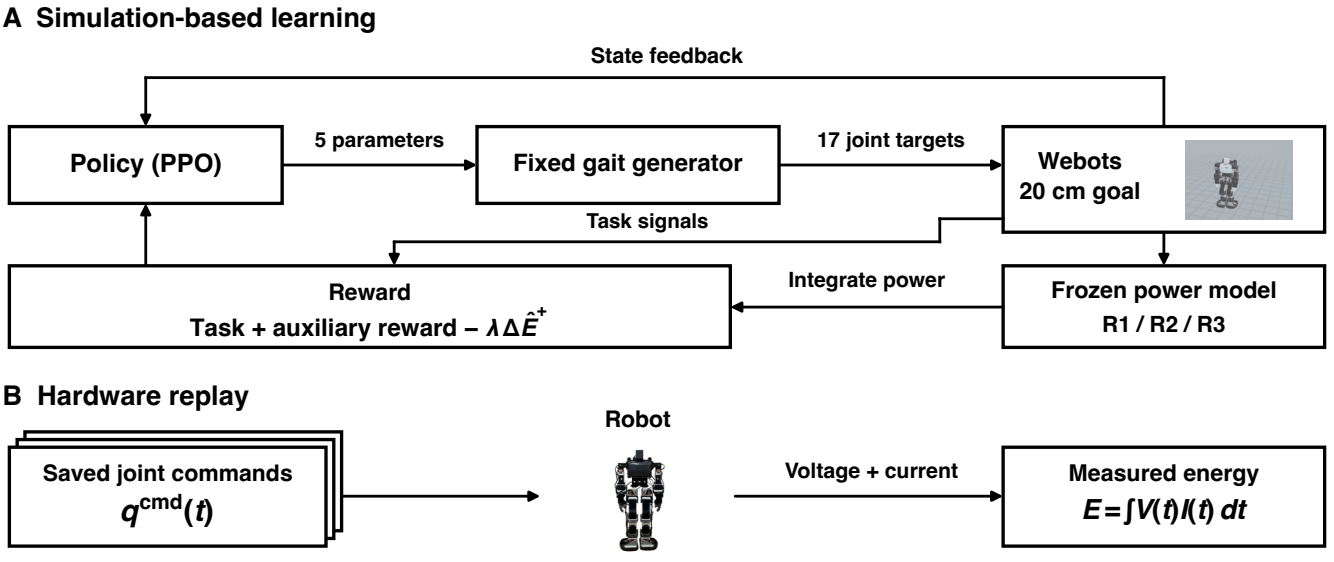
A Simulation-based learning
State feedback
Policy (PPO)
5 parameters
Fixed gait generator
17 joint targets
Webots
20 cm goal
Task signals
Reward
Task + auxiliary reward − λΔÊ+
Integrate power
Frozen power model
R1 / R2 / R3
B Hardware replay
Saved joint commands
q^cmd(t)
Robot
Voltage + current
Measured energy
E = ∫V(t)I(t) dt